

Transformation ↔ matrix

Once a basis is fixed the two languages are interchangeable, and every statement about a map has a matrix statement beside it. The point of reading them in parallel is to notice that neither side is primary — the matrix is not an approximation to the transformation, it is the transformation written in coordinates.

THE CORRESPONDENCE ITSELF

3

1

The map and its matrix
unique $m \times n$ matrix per linear T

$T(\mathbf{x}) = A\mathbf{x}$

Every linear map from $\mathbb{R}^n$ to $\mathbb{R}^m$ is matrix multiplication and nothing else — there is no more general kind of linear map hiding elsewhere. The uniqueness matters as much as the existence: two different matrices cannot represent the same map in the same basis.

2

Columns are images of basis vectors
how the matrix is built

$T(\mathbf{e}_j) = j\text{-th column of } A$

The whole construction in one line. Feed the map each basis vector in turn and record what comes back — the results are the columns. This is why knowing a linear map on a basis is knowing it everywhere.

3

Dimensions
domain $\mathbb{R}^n$, codomain $\mathbb{R}^m$

n columns, m rows

Columns count inputs, rows count outputs. A linear operator has $V = W$, so its matrix is square — which is the only case where eigenvalues, determinants and inverses are even defined.

OPERATIONS TRANSLATE

2

4

Composition
note the order

$S \circ T \leftrightarrow BA$

Applying T then S corresponds to the product BA , not AB — the matrix acting first stands on the right, because it meets the vector first. This is the reason **matrix multiplication** is defined the way it is rather than entry by entry.

5

Identity and inverse
 T bijective $\Leftrightarrow A$ invertible

$I \leftrightarrow I, T^{-1} \leftrightarrow A^{-1}$

The identity transformation is the identity matrix in every basis — the one matrix a change of basis cannot alter. Invertibility transfers exactly, which is what makes the **invertibility equivalence** a statement about transformations too.

SUBSPACES TRANSLATE

2

6

Image and kernel
always

$\text{im}(T) \leftrightarrow \text{Col}(A), \text{ker}(T) \leftrightarrow \text{Null}(A)$

The reachable outputs are the column space; the collapsed inputs are the null space. So rank-nullity is a statement about transformations that happens to be provable by **row reduction**.

7

Isomorphism
 $\dim V = \dim W$, both finite

invertible square matrix

Two spaces of the same finite dimension are isomorphic, and the isomorphism is a change of coordinates. This is why $\mathbb{P}^3$ and $\mathbb{R}^{2 \times 2}$ are the same object twice — both four-dimensional, so both are $\mathbb{R}^4$ in disguise.

The dictionary is exact and it is one-to-one, but only once a basis is chosen. Change the basis and the same transformation gets a different matrix — which is what makes **similarity** the right notion of sameness for matrices, and why properties surviving a basis change are the ones worth naming.